

NFCU DataPower OpenShift Deployment Lab Guide

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Lab Format: Hands-on

Lab goal

In this lab, students deploy IBM DataPower Gateway on OpenShift, expose the web management interface and a Multi-Protocol Gateway route, validate the deployment, and extend the deployment with a Git-based configuration workflow.

Learning outcomes

By the end of this lab, students should be able to:

1. Log in to an OpenShift cluster.
2. Create or select the target project.
3. Install the IBM DataPower Operator.
4. Create the secrets and ConfigMaps required for a base deployment.
5. Deploy a DataPowerService instance.
6. Expose the Web UI (for POT only) and MPG endpoint.
7. Validate that the platform is working.
8. Update the deployment to pull configuration from Git.

Lab inputs

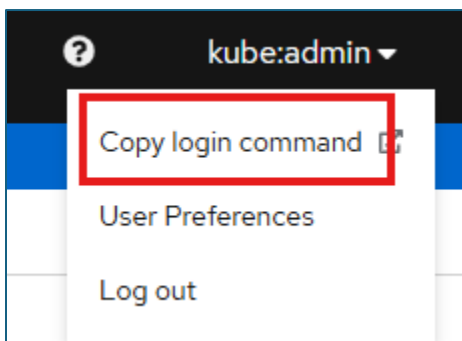
Before you begin, collect the following information from your instructor or lab environment:

Item	Value
OpenShift console URL	<code>https://console-openshift-console.apps.<cluster-name>.<domain></code>

OpenShift API endpoint	<code>https://api.<cluster-name>:6443</code>
OpenShift username	kubeadmin or the username provided to you
OpenShift password or token	Provided by your instructor or administrator
IBM entitlement key	Provided before the lab starts
Email for entitlement secret	Your email address
Cluster apps domain	Determine during the lab if needed

Getting your OpenShift login command

- In the OpenShift console, open the user menu and select **Copy login command**.



- Select **Display Token**.

Example token login command:

Command:

```
oc login --token=<token> --server=https://api.<cluster-name>:6443
```

Files used in this lab

Download the lab files from the GitHub repository before starting: [NFCU_DataPower_MVP1](#).

Files referenced in this lab include:

- `Datapower.yaml`
- `web-mgmt.cfg`
- `webgui-service.yaml`

- `webgui-route.yaml`
- `route.yaml`
- `service.yaml`
- `pull-from-git.cfg`
- `github.com.cer`
- `pot-additionalDomainConfig.yaml`

Prerequisites

Complete the following before the lab begins:

- Access to an OpenShift cluster.
- The `oc` CLI installed and working.
- The IBM entitlement key.
- Access to the GitHub lab repository.
- Permission to create resources in the target namespace.

Lab 1: Log in to OpenShift

Objective

Authenticate the cluster and confirm that CLI access works.

Steps

1. Get your login command from the OpenShift console, or use the username and password provided by your instructor.
2. Run the login command.

Command:

```
oc login --token=<token> --server=https://api.<cluster-name>:6443
```

3. Confirm that authentication succeeded.

Command:

```
oc whoami  
oc cluster-info
```

4. Record the current user and cluster details for your lab notes.

Checkpoint

- `oc whoami` returns your username.
- `oc cluster-info` returns cluster details.

Lab 2: Create or select the project

Objective

Use the target namespace for the operator and DataPower deployment.

Steps

1. Create the project if it does not already exist.

Command:

```
oc new-project nfcu-demo
```

2. If the project already exists, switch to it.

Command:

```
oc project nfcu-demo
```

3. Confirm the active project.

Command:

```
oc project
```

Checkpoint

- The current project is nfcu-demo.

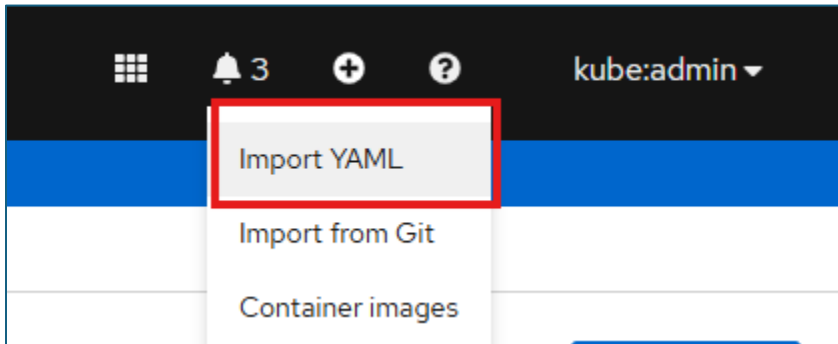
Lab 3: Install the DataPower Operator

Objective

Install the catalog source and the IBM DataPower Operator.

Steps

1. Open **datapower-catalog-source-latest.yaml** and copy its contents.
2. In the OpenShift console, Click the “+” to open **Import YAML** and paste the file contents.



3. Set the namespace to **openshift-marketplace**, then create the resource.
4. Review the created catalog source in the console and confirm it was applied successfully.
5. Go to **Operators > OperatorHub** and search for **IBM DataPower Gateway**.
6. Select the operator and click **Install**.
7. Choose the installation mode, update channel, and approval strategy required for your lab environment.
8. Click **Subscribe** and wait for the installation to complete.
9. Open **Operators > Installed Operators** and confirm the operator appears without errors.

Checkpoint

- The subscription exists.
- The DataPower ClusterServiceVersion is installed.
- The operator pod is running.

Lab 4: Prepare secrets and ConfigMaps

Objective

Create the resources required for the base DataPower deployment.

Steps

1. Confirm that you are in the `nfcu-demo` project.

Command:

```
oc project nfcu-demo
```

2. Create the admin password secret.

Command:

```
oc create secret generic dp-admin-password --from-literal=password=admin -n nfcu-demo
```

3. Create the GitHub certificate secret referenced by `Datapower.yaml`.

Command:

```
oc create secret generic github.com.cer --from-file=github.com.cer -n nfcu-demo
```

4. Create the IBM entitlement secret so the operator can pull the DataPower image from the IBM Container Registry.

Command:

```
oc create secret docker-registry ibm-entitlement-key --docker-server=cp.icr.io --docker-username=cp --docker-password=<YOUR_IBM_ENTITLEMENT_KEY> --docker-email=<YOUR_EMAIL> -n nfcu-demo
```

5. Link the entitlement secret to the default service account for image pulls.

Command:

```
oc secrets link default ibm-entitlement-key --for=pull -n nfcu-demo
```

6. Create the web management ConfigMap from `web-mgmt.cfg`.

Command:

```
oc create configmap web-mgmt-cfg --from-file=web-mgmt.cfg -n nfcu-demo
```

7. Verify that the required resources exist.

Command:

```
oc get configmap web-mgmt-cfg -n nfcu-demo  
oc get secret dp-admin-password -n nfcu-demo  
oc get secret github.com.cer -n nfcu-demo  
oc get secret ibm-entitlement-key -n nfcu-demo
```

Checkpoint

- web-mgmt-cfg exists.
- dp-admin-password exists.
- github.com.cer exists.
- ibm-entitlement-key exists.
- The entitlement key is linked to the default service account.

Lab 5: Deploy DataPower

Objective

Create the DataPowerService resource and wait for it to become ready.

Steps

1. Review `Datapower.yaml` and confirm the namespace, version, deployment name, and domain configuration are correct for your environment.
2. In the OpenShift console, go to **Operators > Installed Operators**.
3. Open **IBM DataPower Gateway**.
4. Under **Provided APIs**, select **DataPowerService**.
5. Click **Create Instance**.
6. In **Form view**, review the key fields such as name, namespace, version, users, domains, and replicas.
7. Switch to **YAML view**, paste the contents of `Datapower.yaml`, and click **Create**.
8. Wait for the DataPowerService status to reach **Running**.
9. Optionally monitor progress from the CLI.

Command:

```
oc get pods -w
oc get datapowerservice quickstart
oc logs quickstart-0
```

Checkpoint

- The pod reaches `Running`.
- The DataPowerService becomes ready.
- Base startup completes successfully.

Lab 6: Expose the web UI and MPG

Objective

Create the services and routes required to access the web UI and the Multi-Protocol Gateway.

Steps

1. Open `webgui-route.yaml`.
2. Update `spec.host` field using the pattern `<route-or-service-name>-<namespace>.<apps-domain>`.
3. If needed, determine the cluster apps domain.
- 4.

Command:

```
oc get ingresses.config/cluster -o jsonpath='{.spec.domain}'
```

Copy output of command in the `<hostname>`

```
! webgui-route.yaml
1  apiVersion: route.openshift.io/v1
2  kind: Route
3  metadata:
4    name: quickstart-webgui
5    namespace: nfcu-demo
6  spec:
7    host: quickstart-webgui-nfcu-demo.<hostname>
8    port:
9      targetPort: webgui-9090
10   tls:
11     insecureEdgeTerminationPolicy: None
12     termination: passthrough
13   to:
14     kind: Service
15     name: quickstart-webgui
16     weight: 100
17   wildcardPolicy: None
18
19 # update host name
20
```

4. Save `webgui-route.yaml` after updating the host name.
5. Apply the web UI service.

Command:

```
oc apply -f webgui-service.yaml
```

6. Apply the web UI route.

Command:

```
oc apply -f webgui-route.yaml
```

7. Apply the MPG service.

Command:

```
oc apply -f service.yaml
```

8. Apply the MPG route.

Command:

```
oc apply -f route.yaml
```

9. List the routes.

Command:

```
oc get route
```

10. Get the web UI route hostname.

Command:

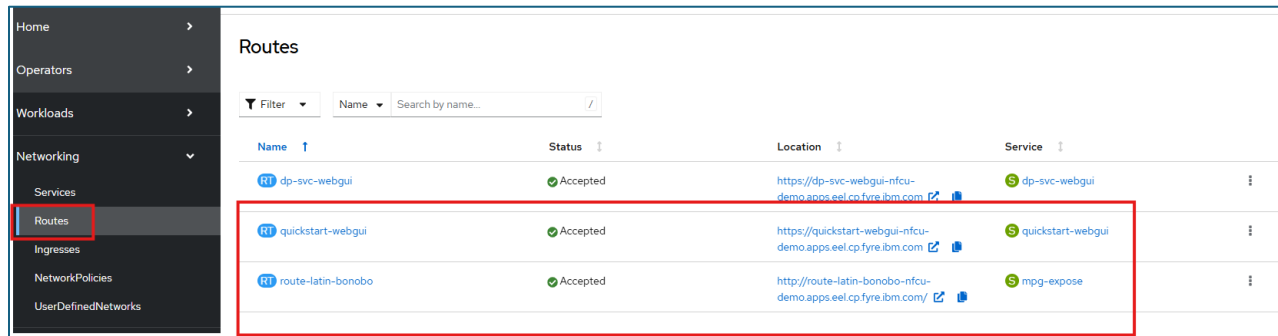
```
oc get route quickstart-webgui -o jsonpath='{.spec.host}'
```

11. Get the MPG route hostname.

Command:

```
oc get route route-latin-bonobo -o jsonpath='{.spec.host}'
```

Note: You can also Validate in the Openshift Console – Networking → Routes



Name ↑	Status ↓	Location ↓	Service ↓
 dp-svc-webgui	Accepted	https://dp-svc-webgui-nfcu-demo.apps.eel.cp.fyre.ibm.com/	 dp-svc-webgui
 quickstart-webgui	Accepted	https://quickstart-webgui-nfcu-demo.apps.eel.cp.fyre.ibm.com/	 quickstart-webgui
 route-latin-bonobo	Accepted	http://route-latin-bonobo-nfcu-demo.apps.eel.cp.fyre.ibm.com/	 mpg-expose

Checkpoint

- The web UI route exists.
- The MPG route exists.

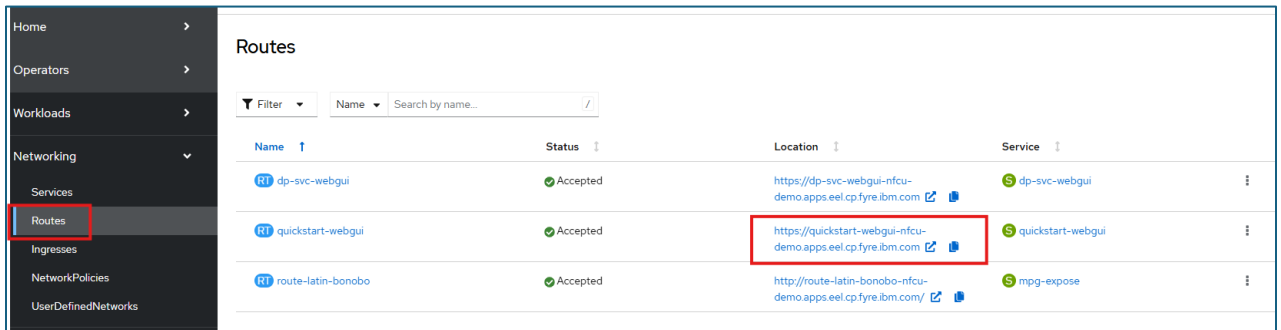
Lab 7: Validate the deployment

Objective

Confirm that the deployment works from both the administrative and API perspectives.

Steps

1. Open the DataPower web UI in a browser using the **quickstart-webgui** route hostname or click on the Location:

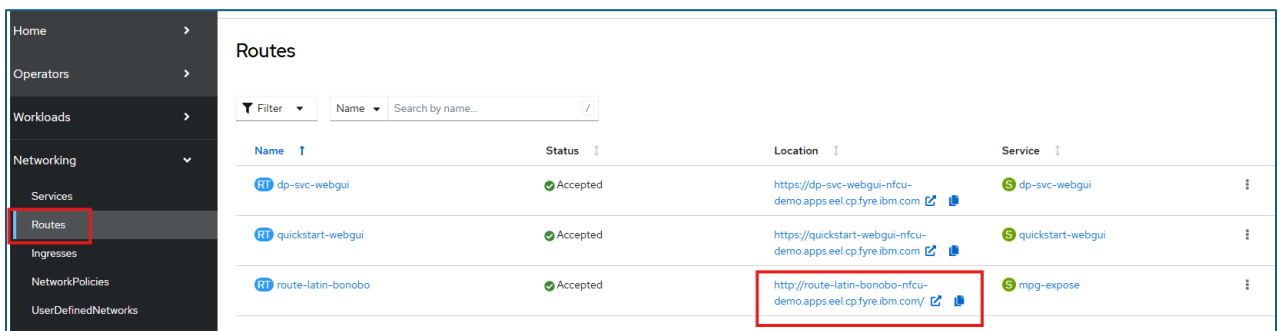


Routes				
Filter Name Search by name...				
Name ↑	Status ↓	Location ↓	Service ↓	
dp-svc-webgui	Accepted	https://dp-svc-webgui-nfcu-demo.apps.eel.cp.fyre.ibm.com	dp-svc-webgui	
quickstart-webgui	Accepted	https://quickstart-webgui-nfcu-demo.apps.eel.cp.fyre.ibm.com	quickstart-webgui	
route-latin-bonobo	Accepted	http://route-latin-bonobo-nfcu-demo.apps.eel.cp.fyre.ibm.com/	mpg-expose	

2. Log in with the lab credentials below.

Field	Value
Username	admin
Password	admin

3. Confirm that the default and demo domains are available.
4. Import [mpg-expose.zip](#) into the demo domain.
5. Select the Link below to Test the mpg-expose MPGW



Routes				
Filter Name Search by name...				
Name ↑	Status ↓	Location ↓	Service ↓	
dp-svc-webgui	Accepted	https://dp-svc-webgui-nfcu-demo.apps.eel.cp.fyre.ibm.com	dp-svc-webgui	
quickstart-webgui	Accepted	https://quickstart-webgui-nfcu-demo.apps.eel.cp.fyre.ibm.com	quickstart-webgui	
route-latin-bonobo	Accepted	http://route-latin-bonobo-nfcu-demo.apps.eel.cp.fyre.ibm.com/	mpg-expose	

6. Confirm that the service returns a JSON response.
7. Review the related OpenShift objects.

Command:

```
oc get pods  
oc get datapowerservice  
oc get route
```

Checkpoint

- Web UI login works.
- The default domain is up.
- The MPG route responds successfully.
- The base platform is ready for the Git-based extension lab.

Lab 8: Pull configuration from Git

Objective

Add a new DataPower domain that uses the GitOps feature to pull configuration from GitHub, then confirm the configuration is visible in the web UI and written back to the repository.

Part A: Create required resources

1. Confirm that you are in the `nfcu-demo` project.

Command:

```
oc project nfcu-demo
```

2. Create the ConfigMap from `pull-from-git.cfg`.

Command:

```
oc create configmap pull-from-git-cfg --from-file=pull-from-git.cfg -n nfcu-demo
```

3. Recreate the GitHub certificate secret to ensure the correct certificate file is being used.

Command:

```
oc delete secret github.com.cer -n nfcu-demo
oc create secret generic github.com.cer --from-file=github.com.cer -n nfcu-demo
```

4. Create the GitHub personal access token secret. Use a token with repository access.
5. IBM Team will provide PAT token.

Command:

```
oc create secret generic pat --from-literal=password=<YOUR_GITHUB_PAT_TOKEN> -n nfcu-demo
```

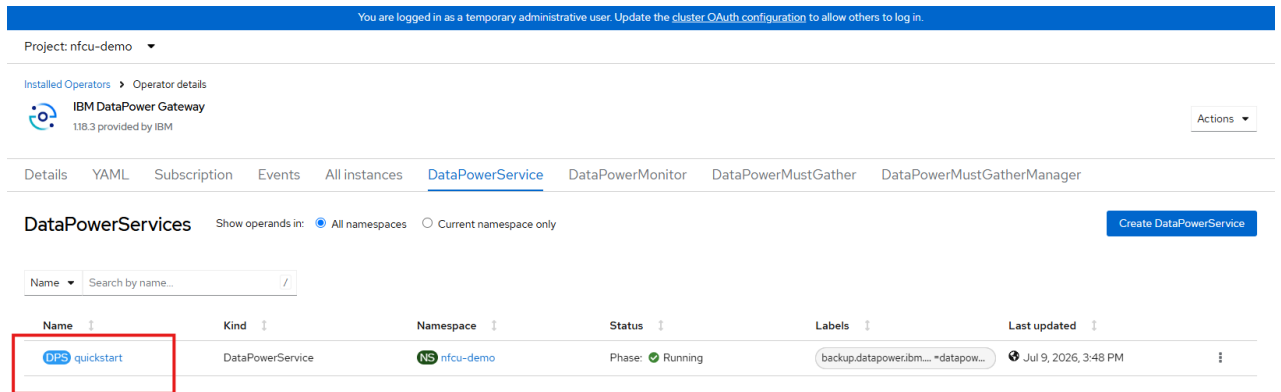
5. Verify the resources.

Command:

```
oc get configmap pull-from-git-cfg -n nfcu-demo
oc get secret github.com.cer -n nfcu-demo
oc get secret pat -n nfcu-demo
```

Part B: Update the DataPowerService

1. In the OpenShift console, open **Operators > Installed Operators > IBM DataPower Gateway > DataPowerService**.
2. Click to open the quickstart DataPowerService.



3. Download the current YAML definition (Download in the Bottom Corner).
4. It will should be named **datapowerservice-quickstart.yaml**
5. Open the (**datapowerservice-quickstart.yaml**) downloaded file in a text editor.
6. Download and open **pot-additionalDomainConfig.yaml** in a text editor from the lab repository.
7. Select All the contents from the **pot-additionalDomainConfig.yaml**
8. Replace the existing **domains**: section in the downloaded DataPowerService YAML (**datapowerservice-quickstart.yaml**) with the updated domain configuration.

Before:

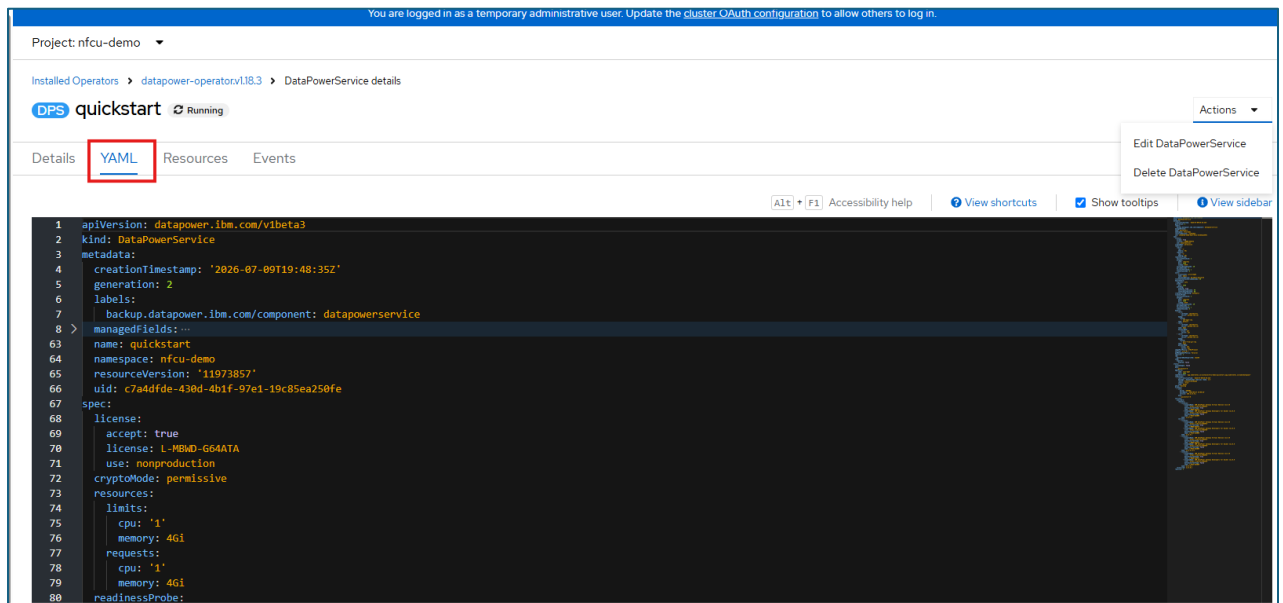
```
114 domains:
115   - certs:
116     - certType: sharedcerts
117       secret: github.com.cer
118     dpApp:
119       config:
120         - web-mgmt-cfg
121     name: default
122   - certs:
123     - certType: sharedcerts
124       secret: github.com.cer
125     name: demo
```

After:

```
114 domains:
115   - certs:
116     - certType: sharedcerts
117       secret: github.com.cer
118     dpApp:
119       config:
120         - web-mgmt-cfg
121     name: default
122   - certs:
123     - certType: sharedcerts
124       secret: github.com.cer
125     passwordMap:
126       - alias: pat
127         secret: pat
128     name: demo
129   - certs:
130     - certType: sharedcerts
131       secret: github.com.cer
132     dpApp:
133       config:
134         - pull-from-git-cfg
135     passwordMap:
136       - alias: pat
137         secret: pat
138     name: Test
```

9. Save the updated YAML (**datapowerservice-quickstart.yaml**).

10. In the OpenShift console select the **YAML** tab of the quickstart DataPowerService



11. Click **Edit DataPowerService**

12. Paste the updated YAML (**datapowerservice-quickstart.yaml**) with the additional configuration

13. Click Save

14. Click Reload

15. Optional Watch the pod restart and return to **Running**.

Command:

```
oc get pods -w
```

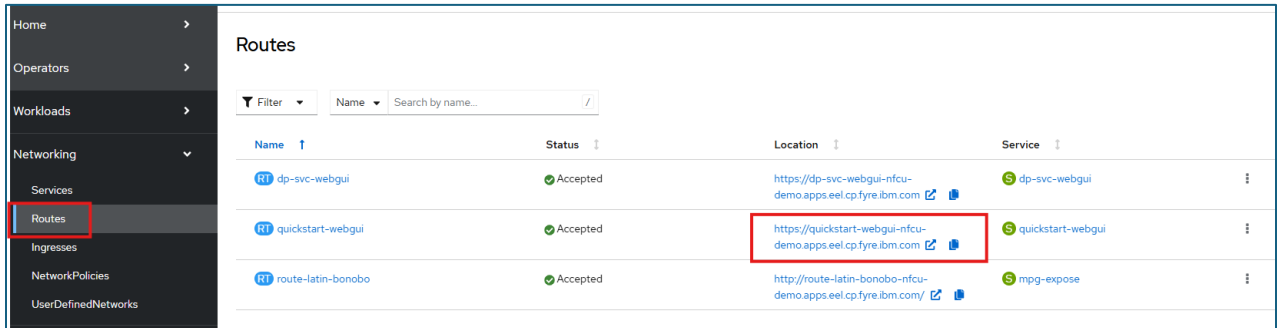
10. Confirm that the updated DataPowerService includes the new Test domain configuration.

Command:

```
oc get datapowerservice quickstart -o yaml
```

Part C: Validate in the web UI

1. Open the DataPower web UI using the quickstart-webgui route.



Routes				
Filter Name Search by name... /				
Name ↑	Status ↓	Location ↓	Service ↓	
dp-svc-webgui	Accepted	https://dp-svc-webgui-nfcu-demo.apps.eel.cp.fyre.ibm.com	dp-svc-webgui	
quickstart-webgui	Accepted	https://quickstart-webgui-nfcu-demo.apps.eel.cp.fyre.ibm.com	quickstart-webgui	
route-latin-bonobo	Accepted	http://route-latin-bonobo-nfcu-demo.apps.eel.cp.fyre.ibm.com	mpg-expose	

2. Log in as **admin** with password **admin**.
3. Confirm that a new Test domain appears alongside default and demo.
4. Switch to the Test domain.
5. Search for the **GitOps** object and open it.
6. Confirm that the repository URL points to **https://github.com/innessrg/NFCU_MVP1_Source_Files.git**.
7. Open **Multi-Protocol Gateway** in the Test domain and review the services pulled from the repository.

Part D: Confirm GitHub

1. Open the repository [lab repository](#) in a browser.
2. Confirm that the expected DataPower domain configuration files are present.

Checkpoint

- pull-from-git-cfg exists.
- github.com.cer exists.
- pat exists.
- The DataPowerService was updated successfully.
- The pod restarted and returned to Running.
- The Test domain appears in the web UI.
- The GitOps object points to the correct repository.

- The expected services appear in the Test domain.
- Configuration is visible in the GitHub repository.

Troubleshooting

If the DataPowerService does not start correctly, use the following checks.

Check pod status and events

```
oc get pods -n nfcu-demo
oc describe pod quickstart-0 -n nfcu-demo
```

Review the **Events** section for pull errors, scheduling failures, or configuration issues.

Check DataPowerService status

```
oc describe datapowerservice quickstart -n nfcu-demo
```

Review the **Status** section for conditions or error messages.

Check operator logs

```
oc logs -n openshift-operators -l name=datapower-operator
```

Use the operator logs to identify reconciliation failures.

Check image pull configuration

If the pod is stuck in ImagePullBackOff or ErrImagePull, verify the entitlement secret and its service account link.

```
oc get secret ibm-entitlement-key -n nfcu-demo
oc get serviceaccount default -n nfcu-demo -o yaml | grep ibm-entitlement-key
```

Check for missing secrets or ConfigMaps

```
oc get secret,configmap -n nfcu-demo
```

Use this command to confirm that all referenced resources from Lab 4 exist.

Check namespace events

```
oc get events -n nfcu-demo --sort-by='.lastTimestamp'
```

Namespace events often provide the fastest explanation for a failed deployment.